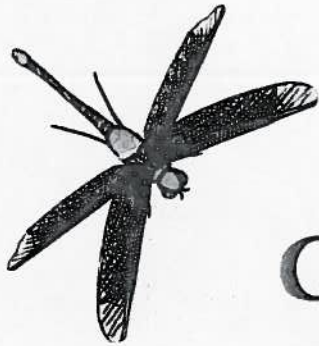




Community Ecology

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Community Ecology's Roots

Every genuine worker in science is an explorer, who is continually meeting fresh things and fresh situations, to which he has to adapt his material and mental equipment. This is conspicuously true of our subject, and is one of the greatest attractions of ecology to the student who is at once eager, imaginative, and determined. To the lover of prescribed routine methods with the certainty of "safe" results, the study of ecology is not to be recommended.

Arthur Tansley, 1923: 97

If we knew what we were doing, it wouldn't be called research.

Albert Einstein

The diversity of life on our planet is remarkable; indeed, among the biggest questions in all of biology are: How did such a variety of life arise? How is it maintained? What would happen if it were lost? Community ecology is that branch of science focused squarely on understanding Earth's biodiversity, including the generation, maintenance, and distribution of the diversity of life in space and time. It is a fascinating subject, but not an easy one. Species interact with their environment and with one another. As we will see in the pages that follow, these interactions underlie the processes that determine biodiversity. Yet, unlike the interacting particles studied by physicists, species also change through time—they evolve. This continual change makes the study of interacting species perhaps even more challenging than the study of interacting particles.

In his 1959 address to the American Society of Naturalists, G. Evelyn Hutchinson posed a simple question: "Why are there so many kinds [species] of animals?" Hutchinson's question remains as fresh and relevant today as it was half a century ago. In this book we will explore what ecologists understand about the processes that drive the distribution of animal and plant diversity across different spatial and temporal scales. In order to appreciate the current state of community ecology, however, it is important to know something about its history, particularly the development of ideas. This first chapter provides a brief summary of that history. Those of you

familiar with the field may skip ahead, while those of you interested in learning more should consult the books and papers by Hutchinson (1978), Colwell (1985), Kingsland (1985), May and Seger (1986), McIntosh (1980, 1985, 1987), and Ricklefs (1987, 2004). Many of these "histories" were written by ecologists actively involved in the field's development, for community ecology is a relatively young science.

What Is a Community?

A **community** is a group of species that occur together in space and time (Begon et al. 2006). This definition is an operational one. Any limits on space and time are arbitrary, as are any limits on the number of species in a community. For example, we might refer to the study of "bird communities" or "fish communities" to delimit the assemblage of interest, recognizing that it is impossible to study all the species that occur together in the same place at the same time. Although most ecologists would be happy with this definition, the concept of what a community is and how it is organized has changed widely through time (Elton 1927; Fauth et al. 1996).

The first community ecologists were botanists who noted what appeared to be repeated associations between plant species in response to spatial and temporal variation in the environment. Frederic Clements, the pioneer of North American plant ecology, viewed these plant associations as a coherent unit—a kind of superorganism—and he presumed that plant communities followed a pattern of succession to some stable climax community (Clements 1916). Limnology, the study of lakes, also adopted a superorganism view of communities. In 1887, the limnologist Stephen Forbes published a famous paper entitled "The lake as a microcosm," in which he stated that all organisms in a lake tend to function in harmony to create a system in balance. Thus, these early ecologists tended to view communities as unique entities, and they became preoccupied with classifying plant and lake communities into specific "types."

This superorganism concept of communities did not sit well with everyone. It was soon questioned by a number of plant ecologists, most notably Henry Gleason (1926) and Arthur Tansley (1939). Gleason asserted that species have distinct ecological characteristics, and that what appear to be tightly knit associations of species on a local scale are in fact the responses of individual species to environmental gradients. Gleason's individualistic concept of communities was ignored at first, but later asserted itself and led to a rejection of Clements's hypotheses in favor of a continuum or gradient theory of plant distributions (Whittaker 1956).

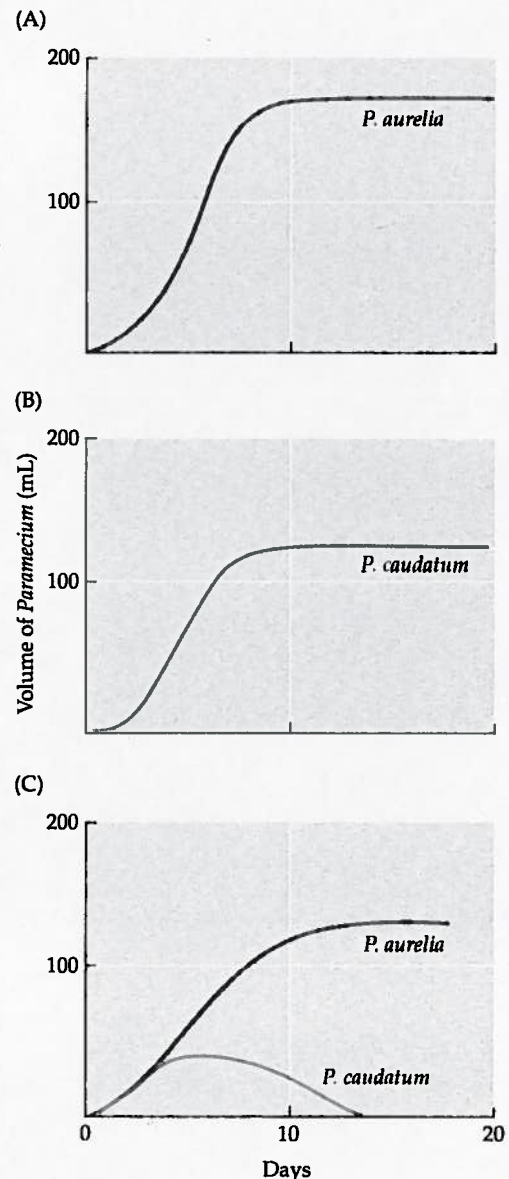
The debate between Clements and Gleason over the nature of communities may seem like a historical footnote today, but at its core is a question that is very much alive: To what extent are local communities—the collections of species occurring together at a site—real entities? Ricklefs (2004) suggested that "ecologists should abandon circumscribed concepts of local communities. Except in cases of highly discrete resources or environments with sharp ecological boundaries, local communities do not exist. What ecologists have called communities in the past should be thought of as point estimates of overlapping regional species distributions." This focus on the interplay

between local and regional processes in determining species associations is a theme that we will return to often in this book.

In contrast to plant ecology, the study of animal communities grew out of laboratory and field studies of populations. Animal population biologists, resource managers, and human demographers were concerned with the factors that regulate the abundance of individuals over time (birth, death, migration). Charles Elton, one of the pioneers of animal community ecology, worked for a time as a consultant for the Hudson's Bay Company, and his thinking was strongly influenced by the fluctuations he observed in the abundance of Arctic animals. Elton was opposed to the "balance of nature" concept espoused by Forbes and others, and in a book entitled *Animal Ecology* (1927), he discussed such important ideas as food webs, community diversity and community invasibility, and the niche. In another book, *The Ecology of Animals*, Elton (1950: 22) proposed that communities have limited membership, stating that in any prescribed area, "only a fraction of the forms that could theoretically do so actually form a community at any one time." Elton went on to note that for animals as well as humans, it appears that "many are called, but few are chosen."

Elton's idea of limited membership was a significant insight, and it meshed well with the concurrent development of mathematical theories of population growth and species interactions. In the 1920s, mathematical ecologists Alfred Lotka (1925) and Vito Volterra (1926) independently developed the now famous equations that bear their names, which describe competition and predation between two or more species. These mathematical models showed that two species competing for a single resource cannot coexist. Gause (1934) experimentally tested this theory with protozoan populations growing in small bottles on a single resource. He found that species grown separately achieved stable densities, but that when pairs of species were grown together, one species always won out and the other species went extinct (Figure 1.1). Other "bottle experiments" with fruit flies, flour beetles, and annual plants produced similar results. The apparent generality of these results led to the formulation of what became known as Gause's **competitive exclusion principle**, which can be stated as "two species cannot coexist on one limiting resource." The competitive exclusion principle had a profound effect on animal ecology at the time and, in a modified form, became a cornerstone of the developing field of community ecology.

Figure 1.1 Results of Gause's competition experiments with two *Paramecium* species (*P. aurelia* and *P. caudatum*) grown separately and together in small containers in the laboratory. (A, B) Each species reached a stable population size (carrying capacity) when grown in isolation. (C) When grown together, however, one species always outcompeted and eliminated the other. (After Gause 1934.)



In the 1940s and 1950s, there was vigorous debate over the competitive exclusion principle and whether populations were regulated by density-dependent or density-independent factors. Important figures in this debate were Elton, Lack, and Nicholson on the side of competition and density-dependent regulation and Andrewartha and Birch on the side of density independence. In 1957, a number of ecologists and human demographers met at the Cold Spring Harbor Institute in Long Island, New York, to debate the issues of population regulation, with little consensus. However, this symposium did lead to one remarkable result. At the end of the published conference proceedings is a paper by G. E. Hutchinson (1957), modestly entitled "Concluding Remarks." In this paper, Hutchinson formalized the concept of the niche and ushered in what I consider the modern age of community ecology.

The Ecological Niche

The concept of the **niche** has a long history in ecology (see Chase and Leibold 2003a for an excellent summary). Grinnell (1917) defined the niche of an organism as the habitat or environment it is capable of occupying. Elton (1927) independently defined the niche as the role a species plays in the community. Gause (1934) made the connection between the degree to which the niches of two species overlap and the intensity of competition between them. Each of these concepts of the niche was incorporated into Hutchinson's thinking when he formalized the niche concept and connected it to the problem of species diversity and coexistence (Hutchinson 1957, 1959). In his "Concluding Remarks," Hutchinson showed how we might quantify an organism's niche, including both biotic and abiotic dimensions of the environment, as axes of an n -dimensional hypervolume (Figure 1.2).

Hutchinson (1957) went on to distinguish between an organism's **fundamental** (or *preinteractive*) niche and its **realized** (or *postinteractive*) niche. The fundamental niche encompasses those parts of the environment a species could occupy in the absence of interactions with other species, whereas the realized niche encompasses those parts of the environment that a species actually occupies in the presence of interacting species (e.g., competitors and predators). In Hutchinson's view, a species' realized niche was smaller than its fundamental niche due to negative interspecific interactions. However, positive interactions between species (mutualisms, commensalisms) can result in a species occupying portions of the environment that were previously unsuitable; in other words, it is possible for the realized niche to be *larger* than the fundamental niche (Bruno et al. 2003). The fact that positive interactions were not explicitly considered in Hutchinson's niche concept shows how completely the ideas of competition and predation permeated ecological thinking at the time.

Hutchinson's definition of the niche provided the framework on which ecologists would build a theory of community organization based on interspecific competition. First, however, they needed to make Hutchinson's concept more workable. An " n -dimensional niche" is fine in the abstract, but empirically, it is impossible to measure an undefined number of niche dimensions. It took one of Hutchinson's students, Robert MacArthur, to

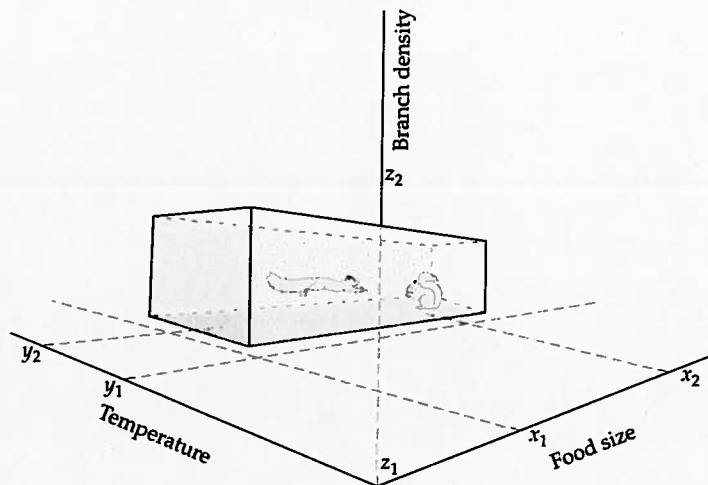


Figure 1.2 Hutchinson's visualization of the niche as an n -dimensional hypervolume. In this hypothetical example, the fundamental niche of a squirrel species is shown along three environmental dimensions. One axis (here labeled y) might define the range of temperatures tolerated by the species, another dimension (x) might describe the range of seed sizes (e.g., acorns) eaten, and a third axis (z) the range of tree branch densities (diameter, volume) that support this squirrel species. Subscripts 1 and 2 represent the upper and lower limits for each niche dimension. (After Hutchinson 1978.)

make the concept operational. MacArthur's approach (1969) was to focus on only a few critical niche axes—those for which competition occurs. If, for example, interspecific competition for seeds limits the number of seed-eating birds in a community, then we should focus our study on some measure of seed availability to define a species' niche (e.g., seed size). Thus, it became possible to examine the distribution of species in "niche space" within a community (see Figure 8.5). More importantly, MacArthur's approach showed how we might quantitatively link overlap in niche space to the process of competitive exclusion (MacArthur 1972).

Shortly after the publication of his "Concluding Remarks" in 1957, Hutchinson (1959) provided another key insight in his published presidential address to the American Society of Naturalists, entitled "Homage to Santa Rosalia, or why are there so many kinds of animals?" Here, Hutchinson pondered a question that goes a step beyond Gause's competitive exclusion principle: If the competitive exclusion principle is true and interspecific competition limits the coexistence of species within the same niche, then how dissimilar must species be in their niches in order to coexist? Hutchinson suggested that the answer might be found in the seemingly regular patterns of difference in body size among members of an ecological guild—co-occurring species that use the same types of resources. He noted that species that were similar in most ways except the sizes of prey eaten tended to differ by a constant size ratio: a factor of about 1.3 in length and 2.0 in body mass. Such regular differences in size among coexisting species

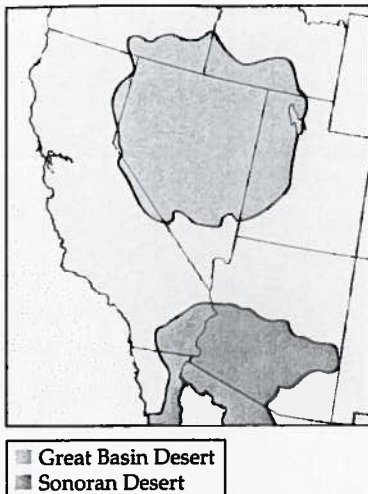
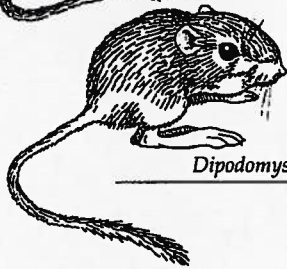


Figure 1.3 Hutchinsonian ratios among desert rodents found in the Great Basin and Sonoran Deserts of the western United States. Differences in body size reflect differences in diet and habitat use (niche differences) between these species. The pattern of body size spacing observed in these two desert rodent ecological guilds is more regular than would be expected by chance. (From Brown 1975.)

Great Basin Desert		Sonoran Desert	
<i>Perognathus longimembris</i>	7.1	<i>Perognathus flavus</i>	
	7.2		
	11.4	<i>Reithrodontomys megalotis</i>	
<i>Microdipodops pallidus</i>	12.5		
	17.1	<i>Perognathus penicillatus</i>	
<i>Peromyscus maniculatus</i>	18.1		
	24.3	<i>Peromyscus maniculatus</i>	
			
<i>Dipodomys merriami</i>	37.6	<i>Dipodomys merriami</i>	
	45.3		
	101		
<i>Dipodomys deserti</i>	102	<i>Dipodomys spectabilis</i>	

were found in many ecological guilds (one example is shown in Figure 1.3) and became known as **Hutchinsonian ratios**.

MacArthur and Levins (1967) built on these ideas and introduced the concept of **limiting similarity**, which specified the minimal niche difference between two competing species that would allow them to coexist. In MacArthur and Levins's theory, species are arrayed linearly along a

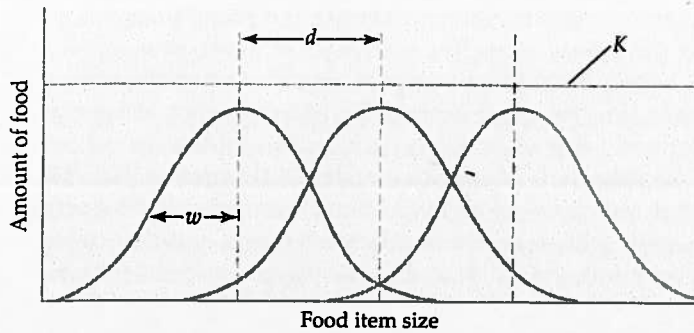


Figure 1.4 The concept of limiting similarity illustrated for three species utilizing a continuum of food resources. K represents a resource continuum (for example, the amount of food as a function of food size). Each species' niche is represented by the mean and the standard deviation (w) of its resource utilization curve, and d is the distance between the mean resource uses of the closest pair of species. MacArthur and Levins (1967) and May and MacArthur (1972) showed that the minimal niche separation required for the coexistence of competing species (under very specific conditions) is $d/w \approx 1$. (After May and MacArthur 1972.)

resource (niche) axis, and each species' resource use is represented by a normal (bell-shaped) utilization curve (Figure 1.4). The overlap between adjacent utilization curves can be used (under specific assumptions) as a measure of the competition coefficients (α 's) in the Lotka-Volterra model of interspecific competition (described in Chapter 7). Using this model of competition, MacArthur and Levins (1967) were able to specify the minimum niche difference required for two species to coexist. Later, May and MacArthur (1972) and May (1973b) used a different approach, based on species in fluctuating environments, to arrive at a very similar outcome: the limiting similarity between two competing species is reached when $d/w \approx 1$, where d is the separation in mean resource use between species and w is the standard deviation in resource use (see Figure 1.4).

Whither Competition Theory?

In less than 50 years, animal community ecology progressed from the simple recognition that species too similar in their niches cannot coexist to the development of a theoretical framework poised to predict the number and types of species found in natural communities based on a functional limit to the similarity of competing species. This was an enormous leap forward, and community ecology seemed well on its way to becoming a more quantitative and predictive science. The heady optimism of the times is reflected in Robert May's (1977a: 195) comment that "the question of the limits to similarity among coexisting competitors is ultimately as deep as the origin of species itself: although undoubtedly modified by prey-predator and mutualistic relations, such limits to similarity are probably the major factor determining how many species there are." In the end, however, the theory of limiting similarity failed to achieve its promise. What happened?

First, there were strong challenges to the idea that interspecific competition is the only, or even the primary, factor structuring communities. Much of the evidence for the importance of interspecific competition in communities was based on descriptive patterns, such as regularly spaced patterns of body size among coexisting species (see Figure 1.3) or “checkerboard” distributions of species on islands (Diamond 1973, 1975). When examined more closely, however, many of these patterns turned out to be indistinguishable from those predicted by models that did *not* include interspecific competition as an organizing force—that is, by null models (Strong et al. 1979; Simberloff and Conner 1981; Gotelli and Graves 1996). Second, predictions of limiting similarity between species turned out to be model-dependent. That is, even though most mathematical models of interspecific competition predict some limit to how similar species may be in their resource use and still coexist, that limit varies widely depending on the assumptions and structure of the model. Thus, as noted by Abrams (1975, 1983b), there is no hard or fixed limit to similarity between species. These challenges caused ecologists to look beyond interspecific competition and to consider the plurality of factors that might determine species diversity. In contrast to the unbridled optimism that characterized community ecology in the 1960s and early 1970s, the next decade was a period of soul searching as ecologists struggled to find a conceptual framework to replace what had seemingly been lost (McIntosh 1987). In the end, however, the idea that communities are organized around strong interspecific interactions was not so much wrong as it was overly simplistic.

New Directions

The “failure” of simple competition-based models to explain community diversity led to important new directions in community ecology, and many of these directions continue to influence how we study ecology. For example, the “null model debate” of the 1970s led directly to an increased emphasis on using field experiments to test ecological hypotheses. Many of these field experiments focused on studying interspecific competition. However, pioneering experiments by Paine (1966), Dayton (1971), and Lubchenco (1978), all working in the marine intertidal zone, also showed that the presence or absence of predators could have dramatic effects on species diversity. These experiments set the stage for a wealth of future work on food webs, trophic cascades, and top-down effects. We will consider these topics in detail in subsequent chapters, as well as more recent approaches to characterizing food webs and other types of ecological networks (see Chapters 10 and 11).

The experimental studies cited above demonstrated how competition and predation may interact to affect species diversity and composition (for example, diversity is increased when predators feed preferentially on a competitive dominant), again setting the stage for subsequent empirical and theoretical work on keystone predation and competition–predation trade-offs. Over time, the accumulation of results from multiple field experiments fostered the application of **meta-analysis** in ecology, in which the outcomes of many experiments are combined and synthesized to arrive at general conclusions (Gurevitch et al. 1992; Osenberg et al. 1999). Today,

ecologists rely heavily on meta-analysis, and there will be many times in this book when we will look to the results of meta-analyses to evaluate the importance of a process in ecology.

The "failure" of a single process (interspecific competition) to account for many of the patterns in species diversity observed in the 1960s and 1970s led ecologists to take a more pluralistic approach to their science (Schoener 1986; McIntosh 1987). A pluralistic ecology recognizes that multiple factors may interact to determine the distribution and abundance of species. The difficulty with pluralism, however, is that it can quickly lead us into a morass. In a recent, thought-provoking article, Vellend (2010: 183) suggested that "despite the overwhelmingly large number of mechanisms thought to underpin patterns in ecological communities, all such mechanisms involve only four distinct kinds of processes: selection, drift, speciation, and dispersal." In Vellend's framework, "selection" encompasses the processes that determine the relative success of species within a local community (e.g., competition, predation, disease), whereas "drift" refers to changes in species' relative abundances due to chance or other random effects, and "dispersal" is the movement of individuals and species into and out of local communities. "Speciation" operates over spatial scales larger than the local community, and it is the process that ultimately generates diversity in regional species pools. Vellend (2010) suggests that conceptual synthesis in community ecology can be achieved by focusing on these four major drivers of species diversity patterns at different spatial and temporal scales (**Figure 1.5**). I agree with Vellend's suggestion. Moreover, I think we are further challenged as community ecologists to illuminate the interior of the "black box" in Figure 1.5, and to better understand how the four basic processes of community ecology interact to determine patterns of biodiversity.

The recognition that local communities bear the footprint of historical and regional processes (Ricklefs and Schluter 1993) is an important insight that grew out of the narrow, local community focus of the 1960s and 1970s. Interestingly, MacArthur (1972) anticipated this paradigm shift, but he

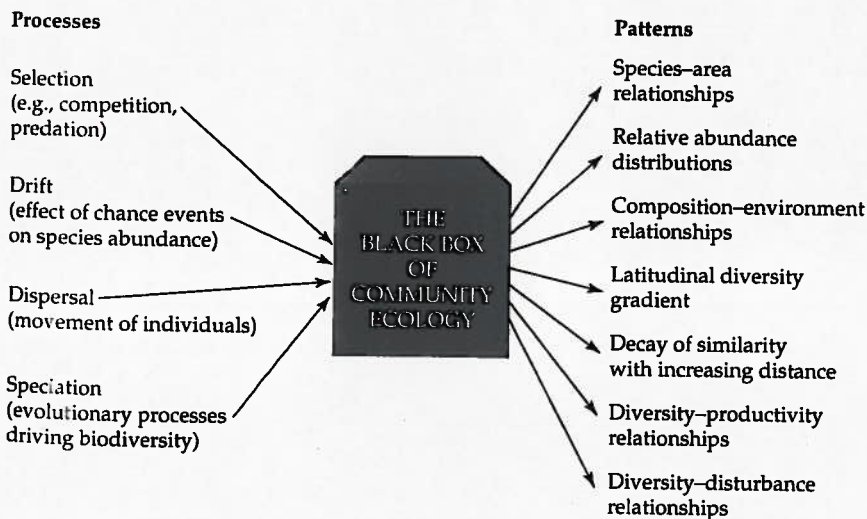


Figure 1.5 A conceptual view of the functioning of community ecology, in which four basic processes (selection, drift, speciation, and dispersal) combine to determine the biodiversity patterns listed on the right. The "black box of community ecology" refers to the fact that there are many ways in which the four processes listed at the left may combine to produce the patterns listed at the right. (After Vellend 2010.)

died too young to be a part of it (see discussion in McIntosh 1987). Simply put, few communities exist in isolation. Instead, the diversity of species within a community is a product of their biotic and abiotic interactions (i.e., species sorting or "selection," along with drift), the dispersal of species between communities, and the composition of the regional species pool (a function of biogeography and evolutionary history). Therefore, we need to consider the processes that regulate diversity at a local scale as well as the processes that link populations and communities into metapopulations and metacommunities and the processes that ultimately generate diversity at regional scales. This is a tall order. In the following chapter, I will use broad-scale diversity gradients, particularly the latitudinal diversity gradient, as a vehicle to begin to think about the processes that generate regional diversity. At geographic scales of regions or continents, biodiversity is a function of evolutionary processes that may play out over millions of years. In addition, chance events in Earth's history can influence a region's size, geomorphology, climate, and the amount of time available for speciation. These historical factors conspire to make our study of the processes that determine regional biodiversity challenging. Of course, there is also little opportunity to do experiments at such vast scales of time and space. But, as we will see, recent advances in molecular biology, phylogenetics, paleontology, and biogeography have greatly facilitated the study of broad-scale diversity patterns, and these new tools are providing the keys to understanding the factors that generate biodiversity at regional scales.